

Instructor(s): Mueller/Korytov

PHYSICS DEPARTMENT

PHY 2049

Exam 3

December 8, 2018

Name (print): _____

Signature: _____

*On my honor, I have neither given nor received unauthorized aid on this examination.***YOUR TEST NUMBER IS THE 5-DIGIT NUMBER AT THE TOP OF EACH PAGE.****DIRECTIONS**

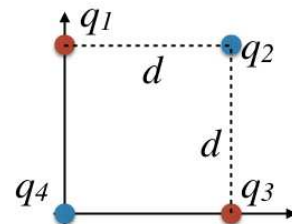
- (1) **Code your test number on your answer sheet (use 76–80 for the 5-digit number).** Code your name on your answer sheet. **Darken circles completely (errors can occur if too light).** Code your student number on your answer sheet.
- (2) Print your name on this sheet and sign it also.
- (3) Do all scratch work anywhere on this exam that you like. At the end of the test, this exam printout is to be turned in. No credit will be given without both answer sheet and printout with scratch work.
- (4) Work the questions in any order. Incorrect answers are not taken into account in any way; you may guess at answers you don't know.
- (5) If you think that none of the answers is correct, please choose the answer given that is closest to your answer.
- (6) **Blacken the circle of your intended answer completely, using a number 2 pencil.** Do not make any stray marks or the answer sheet may not read properly. Completely erase all incorrect answers, or take a new answer sheet.
- (7) As an aid to the examiner (and yourself), in case of poorly marked answer sheets, please circle your selected answer on the examination sheet. Please remember, however, that in the case of a disagreement, the answers on the bubble sheet count, **NOT** what you circle here. Good luck!!!

>>>>>>>>**WHEN YOU FINISH**<<<<<<<<

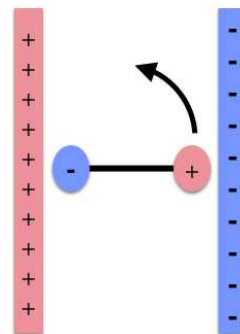
Hand in the answer sheet separately.

Constants		
$\mu_B = 9.27 \times 10^{-24} \text{ J/T}$	$m_p = 1.67 \times 10^{-27} \text{ kg}$	$m_e = 9.11 \times 10^{-31} \text{ kg}$
$e = 1.6 \times 10^{-19} \text{ C}$	$k = 9 \times 10^9 \text{ N m}^2/\text{C}^2$	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$
$\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$	$c = 3 \times 10^8 \text{ m/s}$	$g = 9.8 \text{ m/s}^2$
$\text{f} = \text{femto} = 10^{-15}$	$\text{p} = \text{pico} = 10^{-12}$	$\text{n} = \text{nano} = 10^{-9}$
$\mu = \text{micro} = 10^{-6}$	$\text{M} = \text{mega} = 10^6$	$\text{G} = \text{giga} = 10^9$

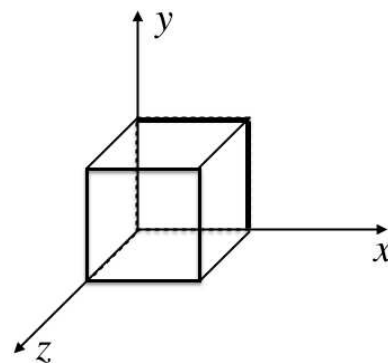
1. Four point charges q_1 to q_4 are placed at the corners of a square as shown in the figure. The charges $q_1 = q_3 = 17\mu\text{C}$. What is the charge of q_2 if the net force on q_4 is zero?



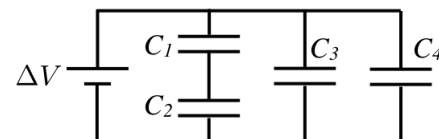
- (1) $-48\mu\text{C}$ (2) $-34\mu\text{C}$ (3) $-67\mu\text{C}$ (4) $-92\mu\text{C}$ (5) $-73\mu\text{C}$
2. A 10 cm long dumbbell has positive and negative charges attached to its ends. Each charge has a magnitude of $Q = 14\mu\text{C}$. This electric dipole is placed between two very large capacitor plates which are separated by 20 cm. The voltage difference across the two plates is 700 V. An external torque is now applied to rotate the dumbbell by 180° . By how much did the potential energy (in mJ) increase?



- (1) 10
(2) 28
(3) 7
(4) 14
(5) 37
3. A cube with an edge length of 2 m is aligned with the axis of our coordinate system with the origin placed at one corner of the cube as shown in the figure. An electric field measured in N/C of the form $\vec{E} = -4y\hat{j} + 7\hat{k}$ is also present. What is the total charge inside the cube?



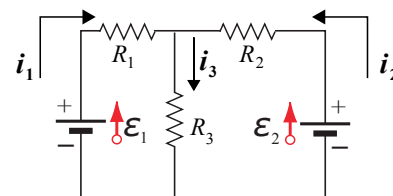
- (1) -283 pC
(2) -94 pC
(3) 85 pC
(4) -9.3 pC
(5) 470 pC
4. Seen from the battery, what is the equivalent capacitance, in μF , of the circuit shown if each capacitor has a capacitance of $2\mu\text{F}$?



- (1) 5 (2) 3 (3) 8 (4) $2/5$ (5) 2
5. A resistor dissipates 0.5 W when a potential difference of 4.0 V is applied. When a current of 0.25 A flows through the resistor, how much power is then dissipated?

- (1) 2 (2) 0.5 (3) 8 (4) 4 (5) 0.25

6. Find the magnitude of the difference of potentials across resistor R_3 in the circuit shown in the figure ($\mathcal{E}_1 = 5\text{ V}$, $\mathcal{E}_2 = 10\text{ V}$, $R_1 = 4\Omega$, $R_2 = 2\Omega$, $R_3 = 1\Omega$).

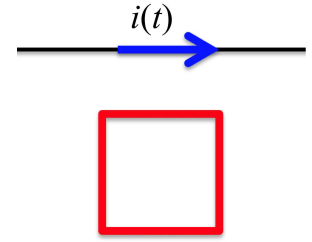


- (1) 3.6 V
(2) 9.4 V
(3) 4.3 V
(4) 2.1 V
(5) 15 V

7. One wire is aligned with the x-axis and carries a current of 2A in the positive x-direction. A second wire is aligned with the y-axis and carries a current of 1A in the positive y-direction. What is the magnetic field at $(x, y) = (2, 1)$, where the distance numbers are given in meters?

(1) 3.0×10^{-7} T (2) 1.5×10^{-7} T (3) 5.0×10^{-7} T (4) 4.0×10^{-7} T (5) 0

8. A square loop made of conductive material is placed near a wire carrying a current as shown in the figure. The current in the wire flows to the right and decreases. What is the direction of the net force on the loop?

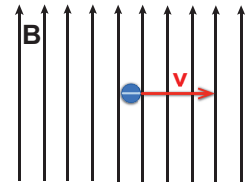


(1) up
(2) down
(3) right
(4) left
(5) the net force is zero

9. An RLC-circuit ($R = 50 \Omega$, $L = 100$ mH, $C = 20 \mu\text{F}$,) is connected to an alternating emf $\mathcal{E} = (100 \text{ V})\sin(\omega_d t)$, where $\omega_d = 500 \text{ s}^{-1}$. What is the amplitude of the current oscillating in the circuit?

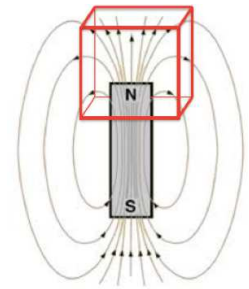
(1) 1.4 A (2) 6.0 A (3) 3.3 A (4) 2.1 A (5) 4.2 A

10. An electron (negative charge!) moves to the right in magnetic field B pointing up, as shown in the figure. What is the direction of the force experienced by the electron?



(1) into the page (2) down (3) up (4) out of page (5) force is zero

11. The north pole of a magnet is enclosed by a Gaussian surface in the form of a cube. The magnetic flux is 2 Tm^2 through the top face of the cube and 1.5 Tm^2 through each of the four side faces. What is the flux (in Tm^2) through the bottom face?



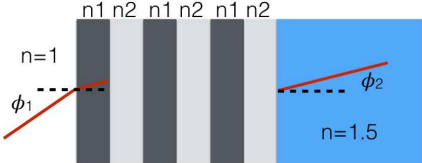
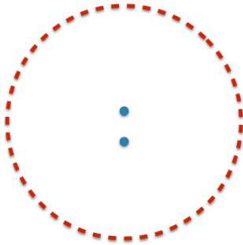
(1) -8
(2) 8
(3) -4
(4) 4
(5) 0

12. A 150W laser beam has a uniform intensity distribution across an area with radius $r = 3$ mm. This beam is directed towards a semi-transparent mirror which reflects 75% of the intensity and transmits 25%. What is the radiation pressure in N/m^2 on this mirror in the area where the beam hits the mirror?

(1) 0.027 (2) 1.3 (3) 0.35 (4) 0.096 (5) 0.73

13. An unpolarized beam of light has intensity I_0 . It is incident on two ideal polarizing sheets. The angle between the axes of polarization of these sheets is θ . Find θ if the emerging light has intensity $I_0/4$:

(1) 45° (2) 30° (3) 60° (4) 15° (5) 75°

14. An inverted image is formed 60 cm from an object by a thin lens located between the two. The image is $1/3$ the height of the object. What is the distance from the object to the lens?
- (1) 45 cm (2) 48 cm (3) 80 cm (4) 30 cm (5) 15 cm
15. A person cannot see clearly objects closer than 1.75 m from her/him. What is the refractive power (in diopters) of the correction lenses that will allow her/him to clearly see objects located 25.0 cm, but not closer, from her eyes? Ignore the distance between her/his eyes and the lenses.
- (1) 3.43 (2) 2.83 (3) 2.78 (4) 3.04 (5) 3.27
16. A glass prism ($n = 1.5$) is coated with 3 parallel stacks of alternating layers of high ($n = 1.90$) and low index ($n = 1.30$) material, each having a thickness of 250 nm. If a laser beam travelling first in vacuum ($n=1$) hits the first layer under an angle of $\phi_1 = 45^\circ$ (see figure), what is the angle ϕ_2 of the beam inside the prism?
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- (1) 28° (2) 45° (3) 67° (4) 15° (5) 35°
17. Two coherent radio-frequency point sources separated by $d = 2.5$ m radiate in phase at a wavelength $\lambda = 0.75$ m. A radio wave detector moves in a circular path of some radius $R > d$ around the mid-point between the two sources (see figure) and measures the radio-wave intensity of the signal. Find how many maxima it detects.
- 
- (1) 14
(2) 8
(3) 10
(4) 2
(5) 6
18. A lens is made from a material with an index of refraction of 1.5. It is coated with a thin film material to reduce the reflection of light of $\lambda = 1064$ nm. The index of refraction of the film material is 1.7. Which layer thickness in nm would minimize the reflectivity at that wavelength.
- (1) 312 (2) 532 (3) 156 (4) 1064 (5) 768
19. Sunlight is incident on a diffraction grating with 200 rulings/mm. What is the highest order, m , that a complete visible spectrum $400 \text{ nm} < \lambda < 700 \text{ nm}$ can be seen?
- (1) 7 (2) 5 (3) 4 (4) 3 (5) 6
20. The next generation of large telescopes will include a 30 m telescope on top of Mauna Kea in Hawaii. If this telescope is pointed towards two identical stars which orbit around each other at the Jupiter-Sun distance of 775 million km, what is the maximum distance d between the telescope and the two stars that the telescope can still resolve them? Assume that the telescope observes light with a wavelength of 500 nm.
- (1) $3.8 \times 10^{19} \text{ m}$ (2) $9.1 \times 10^{20} \text{ m}$ (3) $1.3 \times 10^{24} \text{ m}$ (4) $4.7 \times 10^{22} \text{ m}$ (5) $8.4 \times 10^{16} \text{ m}$